

EXP17: Prolog Program to Sum the Integers from 1 to N

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Aim:

To develop a Prolog program that computes the sum of integers from 1 to N using recursion.

Algorithm:

Step 1: Start the program.

Step 2: Accept the value of N.

Step 3: Initialize the base condition where the sum of 0 integers is 0.

Step4: Check whether N is greater than 0.

Step 5: Reduce the value of N by 1 and recursively calculate the sum of previous integers.

Step 6: Add the current value of N to the obtained sum.

Step 7: Continue the process until N becomes 0.

Step 8: Display the final sum.

Step 9: Stop the program.

Source Code:

% Program to sum the integers from 1 to N using recursion

sum(0, 0).

sum(N, Sum) :-

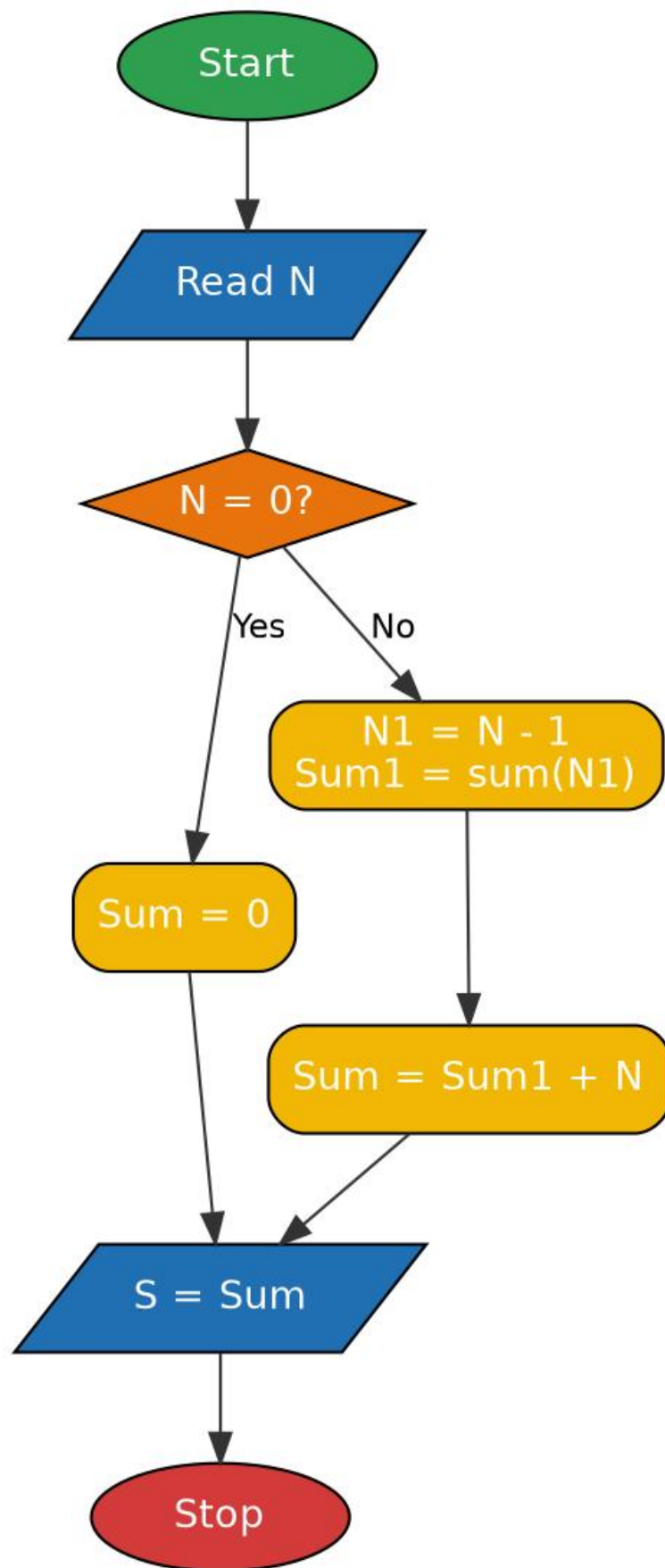
 N > 0,

 N1 is N - 1,

 sum(N1, Sum1),

 Sum is Sum1 + N.

Flowchart:



Output:

```
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
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For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?- cd('C:/Users/yanar/Documents/CSA1717').
true.

?- [sum].
true.

?- sum(10, Sum).
Sum = 55
```

Result:

Thus the Prolog program to compute the sum of integers from 1 to N was written, executed successfully and the output was verified.